3. Functional Requirements

A. Process Analyst Requirements

FR-01: Process analysts can create a new business process by entering its name and description.

FR-02: Process analysts can view a list of saved business processes.

FR-03: Process analysts can open a saved process to view its activities and dependencies.

FR-04: Process analysts can edit the name and description of an existing process.

FR-05: Process analysts can add a new activity to a process.

FR-06: Process analysts can edit an activity's name, description, and type.

FR-07: Process analysts can delete an activity that is no longer required.

FR-08: Process analysts can define activity types, such as task, decision, approval, and output.

FR-09: Process analysts can connect two activities by creating a dependency.

FR-10: Process analysts can edit or remove an existing dependency.

FR-11: Process analysts can define starting and ending activities for a process.

FR-12: Process analysts can save a completed process model in the database.

B. Process Import and Validation Requirements

FR-13: Process analysts can import a process definition from a supported JSON file.

FR-14: System validation checks imported files before saving process data.

FR-15: System validation displays an error message when an imported file contains missing or invalid fields.

FR-16: Process analysts can export a saved process definition as a JSON file.

FR-17: System validation checks whether all dependencies refer to existing activities.

FR-18: System validation detects circular dependencies and displays a warning.

C. Change Scenario Requirements

FR-19: Process analysts can create a change scenario for a selected process.

FR-20: Process analysts can select an activity, decision condition, approval requirement, or dependency to change.

FR-21: Process analysts can enter a proposed change and its new value or condition.

FR-22: System storage saves a proposed change as a separate scenario without overwriting the original process.

FR-23: Process analysts can view a list of saved change scenarios.

FR-24: Process analysts can edit or delete a saved scenario.

FR-25: Process analysts can run impact analysis for a selected change scenario.

D. Dependency Analysis Requirements

FR-26: System graph processing represents process activities and dependencies as a directed graph.

FR-27: System analysis identifies activities directly connected to a selected activity.

FR-28: System analysis identifies all reachable downstream activities affected by a selected change.

FR-29: System analysis identifies paths connecting the changed activity to downstream activities.

FR-30: System analysis distinguishes directly connected activities from indirectly connected activities.

FR-31: System analysis identifies whether a changed activity affects an approval, decision, or final output.

FR-32: Process analysts can view identified affected activities in a readable list.

E. Process Simulation Requirements

FR-33: Process analysts can run a simulation using the original process definition.

FR-34: Process analysts can run a simulation using a modified scenario.

FR-35: Simulation processing follows process conditions to determine which activities are executed.

FR-36: System comparison identifies differences between the original and modified execution paths.

FR-37: System comparison identifies activities executed in the original process but skipped in the modified process.

FR-38: System comparison identifies activities newly required by the modified process.

FR-39: System comparison identifies changes in final outputs produced by both simulations.

FR-40: Process analysts can view a comparison of original and modified simulation results.

F. Visualization and Reporting Requirements

FR-41: Process analysts can view a process dependency graph.

FR-42: Graph visualization highlights affected activities after impact analysis.

FR-43: Process analysts can view original and modified process paths for comparison.

FR-44: Process managers can view completed impact analysis results.

FR-45: Process managers can view a summary of affected activities, approval steps, and final outputs.

FR-46: Process managers can export analysis results as a report.

G. System Administration Requirements

FR-47: System administrators can view stored process models and change scenarios.

FR-48: System administrators can delete test process data that is no longer required.

FR-49: System administrators can view application errors recorded during process validation or simulation.

FR-50: System displays a confirmation message after successfully saving a process, scenario, or report.